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THE AI BRIDGE — WEBSITE IMPLEMENTATION PLAN (PHASE-WISE)

Hum Hain Pakistan | SDG Youth Campaign | IIUI (FMS)

Version: 1.0

Stack: Next.js + Tailwind + shadcn/ui | Supabase | Vercel | SMTP

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PURPOSE OF THIS DOCUMENT

This file is the single source of truth for building "The AI Bridge" webapp.

It contains 5 implementation phases, each with clear goals, tasks, deliverables,

and a "done when" checklist. Hand this to Antigravity / Gemini / your dev team.

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SECTION 0 — PROJECT SUMMARY (WHAT WE ARE BUILDING)

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WHAT IT IS:

An internal management dashboard + public awareness site for The AI Bridge

project, with a student self-service certificate portal.

WHO USES IT:

- ADMIN : Full control. Manages members, permissions, events, settings.

- MEMBER : Working team (10 SDG Ambassadors). Permissions toggled by Admin.
- USER : View-only (students + public visitors).

CORE FLOW:

1. Student fills PRE-SURVEY (Google Form) -> row lands in Google Sheet.
2. Student fills POST-SURVEY (Google Form) -> row lands in Google Sheet.
3. A Supabase cron job pulls the sheet every few minutes -> stores Name, Email, Student ID in Supabase.
4. System matches PRE + POST by (Student ID + Email + Full Name).
5. If BOTH matched -> Certificate becomes "Ready".
   If ONLY one filled -> shows "In Progress" + a Member can click "Send Post-Survey Reminder" (email with the post form link).
6. Member/Admin clicks "Send Certificate" -> PDF generated (with QR code) -> emailed via SMTP (PDF attached + download link).
7. Anyone can scan the QR -> opens public verify page -> shows VALID.
8. Student can log in (Google) -> see status -> download own certificate.
9. Admin/Members upload Events with photos -> shown in public Gallery.

KEY DECISIONS LOCKED:

- Google Sheet is READ-ONLY (we only pull Name + Email + Student ID).
- No manual certificate revoke — validity is purely based on form matching.
- Login: Google Sign-In for ALL roles, handled 100% by Supabase Auth.
- SMTP is only for sending reminder + certificate emails (not login).
- Permissions are set member-by-member, apply instantly.
- Event visibility (public/internal) decided per event by Admin.
- Dashboard shows maximum useful metrics with clean UX/UI.

- Branding: theme generated from panaflex / posters / logo in the assets folder.

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SECTION 1 — FINAL TECH STACK

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FRONTEND

- Next.js (App Router)
- Tailwind CSS
- shadcn/ui components
- Hosted on Vercel

BACKEND / DB / AUTH / STORAGE

- Supabase (all-in-one):
  * Postgres DB
  * Supabase Auth (Google OAuth)
  * Supabase Storage (event images + certificate PDFs)
  * Supabase Edge Functions (cron pull + email sending)
  * Row Level Security (RLS) for role-based access

CERTIFICATE GENERATION

- Server-side HTML -> PDF (PDFKit or Puppeteer/Playwright)
- QR code embedded -> links to /verify/[certificate-id]

EMAIL

- Nodemailer over SMTP

- Sender: Gmail App Password (fallback) or Brevo (official sender)

- Sends: Post-Survey Reminder + Certificate delivery

GOOGLE SHEET SYNC

- Published CSV / "anyone with link can view"

- Supabase Edge Function + cron (every 1-5 minutes)

- One-way: Google -> Supabase (read-only, upsert)

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SECTION 2 — ROLES & PERMISSIONS MATRIX

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Feature / Action	Admin	Member (toggle)	User
View dashboard	Yes	Yes	No
View students list (pre/post status)	Yes	Yes	No
Send Post-Survey Reminder email	Yes	TOGGLE	No
Send Certificate email	Yes	TOGGLE	No
Upload Event (with pictures)	Yes	TOGGLE	No
Manage Members & permissions	Yes	No	No
Set event public/internal	Yes	TOGGLE	No
View public Event Gallery	Yes	Yes	Yes
Student portal (own cert status)	Yes	Yes	Yes
Verify page (QR destination)	Yes	Yes	Yes

RULES:

- Permissions are set MEMBER-BY-MEMBER (not by team).
- Admin toggles each permission individually.
- Toggle changes apply INSTANTLY (no re-login).

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SECTION 3 — WEBSITE PAGES (ROUTE LIST)

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PUBLIC (no login)

1. / -> Landing / Home
   (intro, SDG mapping, team, recent events,
   CTA: Take Pre-Survey / Take Post-Survey)
2. /events -> Public Events Gallery
3. /verify/[certificate-id] -> Certificate verification page (QR target)
4. /about -> Project details, leadership, SDG mapping

AUTH

5. /login -> Google Sign-In (Supabase Auth)

LOGGED-IN (any role)

6. /portal -> Student Portal
   (auto-match by Google email; fallback asks
   for Student ID; shows status + download)

LOGGED-IN (Admin + Members)

7. /dashboard -> Metrics overview

8. /students -> Students list + action buttons

(buttons visible only if permission ON)

9. /events/manage -> Create / edit / delete events, upload images

10. /certificates -> Certificates log (who, when, which workshop)

ADMIN ONLY

11. /admin/members -> Add/remove members, toggle permissions

12. /admin/settings -> SMTP config, form URLs, certificate settings

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SECTION 4 — DATABASE SCHEMA (SUPABASE / POSTGRES)

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TABLE: profiles

id uuid PK (references auth.users)

email text

full_name text

role text CHECK (role IN ('admin','member','user'))

created_at timestamptz

TABLE: member_permissions

member_id uuid PK (references profiles.id)

can_send_reminder boolean DEFAULT false

can_send_cert boolean DEFAULT false

can_upload_event boolean DEFAULT false

can_set_visibility boolean DEFAULT false

updated_at timestamptz

TABLE: survey_responses

id uuid PK

student_id text

full_name text

email text

survey_type text CHECK (survey_type IN ('pre','post'))

submitted_at timestamptz

raw_row jsonb

UNIQUE (student_id, email, survey_type)

TABLE: students (derived / matched view or table)

id uuid PK

student_id text UNIQUE

full_name text

email text

pre_filled boolean

post_filled boolean

status text -- 'in_progress' | 'ready' | 'certificate_sent'

updated_at timestamptz

TABLE: certificates

id uuid PK

certificate_id text UNIQUE -- e.g. AIB-2025-0001

student_id text

full_name text

email text

workshop_name text

completion_date date

pdf_url text -- Supabase Storage path

issued_by uuid -- profiles.id

issued_at timestamptz

email_sent boolean

email_sent_at timestamptz

TABLE: events

id uuid PK

title text

event_date date

location text

description text

cover_image_url text

gallery_urls text[] -- array of Supabase Storage paths

sdg_tags text[] -- e.g. ['SDG4','SDG10']

team_tag text -- 'awareness' | 'workshops' | 'analyzing'

visibility text CHECK (visibility IN ('public','internal'))

created_by uuid

created_at timestamptz

TABLE: email_log

id uuid PK

type text -- 'reminder' | 'certificate'

recipient_email text

student_id text

sent_by uuid

status text -- 'sent' | 'failed'

error text

created_at timestamptz

TABLE: settings

key text PK

value jsonb

updated_at timestamptz

-- stores: SMTP config, form URLs, cert template config, signatory info

STORAGE BUCKETS:

event-images/ -> public (event cover + gallery)

certificates/ -> private (served via signed URLs)

RLS (Row Level Security) — high level:

- profiles: users read own; admin reads/writes all

- member_permissions: admin only writes; member reads own

- survey_responses/students: admin + permitted members only

- certificates: student reads own; admin/members read all

- events: public events readable by all; internal only by admin/members

- settings: admin only

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SECTION 5 — THE 5 IMPLEMENTATION PHASES

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PHASE 1 — FOUNDATION & SETUP

GOAL: Project scaffolding, accounts, database, auth working end-to-end.

TASKS:

1.1 Create Next.js (App Router) project + Tailwind + shadcn/ui.

1.2 Create Supabase project. Save URL + anon key + service role key.

1.3 Run all SQL from SECTION 4 (tables, RLS, storage buckets).

1.4 Configure Supabase Auth -> enable Google OAuth.

- Google Cloud Console -> OAuth credentials -> redirect URI to Supabase.

1.5 Set up Vercel project + link to GitHub repo.

1.6 Build global layout: header, footer, nav (role-aware), theme tokens from branding assets (panaflex/poster/logo).

1.7 Build /login page with Google Sign-In. On first login, auto-create a row in profiles with role='user'.

1.8 Seed an Admin account manually (SQL update role='admin').

1.9 Create middleware to protect role-based routes.

DELIVERABLES:

- Live Next.js app on Vercel (blank but styled).
- Supabase project fully set up.
- Google login working for all users.
- Admin account seeded.

DONE WHEN:

[ ] Any Google account can log in.

[ ] First-time login creates a profile row.

[ ] Admin sees admin-only routes; user does not.

[ ] Branding colors/fonts applied site-wide.

PHASE 2 — DATA PIPELINE (GOOGLE SHEET -> SUPABASE)

GOAL: Survey responses flow from Google Forms into Supabase automatically.

TASKS:

2.1 Confirm Google Form fields: Student ID, Full Name, Email.

2.2 Publish the linked Google Sheet as CSV ("anyone with link can view").

2.3 Write Supabase Edge Function: fetch sheet CSV -> parse -> upsert into survey_responses (UNIQUE on student_id + email + survey_type).

2.4 Schedule the Edge Function via Supabase cron (every 1-5 min).

2.5 Build the "matching" logic:

- Group rows by (student_id + email + full_name).
- If pre exists and post exists -> status='ready'.
- If only one exists -> status='in_progress'.
- Write/update the students table.

2.6 Build the /students page (Admin + Members):
- Table: Name, Student ID, Email, Pre?, Post?, Status.
- Filter by status.
- Show "Send Reminder" and "Send Certificate" buttons ONLY if the logged-in member has the matching permission toggled ON.

2.7 Build the /dashboard with live metrics pulled from Supabase.

DELIVERABLES:
- Automated sync from Google Sheet.
- Working students table with correct statuses.
- Dashboard showing real numbers.

DONE WHEN:
[ ] Submitting a Google Form -> row appears in Supabase within 5 min.
[ ] Both-forms-filled student correctly shows 'ready'.
[ ] One-form-only student correctly shows 'in_progress'.
[ ] Permission-gated buttons hide/show correctly per member.

PHASE 3 — EMAILS, CERTIFICATES & QR VERIFICATION

GOAL: The full certificate lifecycle works: remind -> generate -> send -> verify.

TASKS:

3.1 Configure SMTP (Gmail App Password or Brevo). Store in settings table.

3.2 Build "Send Post-Survey Reminder" action:

- Emails the student the post-survey link.
- Logs in email_log.

3.3 Build certificate PDF generator (Edge Function or Next API route):

- HTML template -> PDF (PDFKit or Puppeteer).
- Fields: Student Name, Student ID, Workshop/Session Name,
  Date of Completion, Project Name
  ("The AI Bridge — Hum Hain Pakistan SDG Youth Campaign"),
  Signatory (Dr. Abdul Zahid Khan + designation),
  Certificate ID (e.g. AIB-2025-0001),
  QR code linking to /verify/[certificate-id].
- Add Sir Zahid's signature image.
- Save PDF to Supabase Storage (certificates bucket).

3.4 Build "Send Certificate" action:

- Generates PDF (if not already).
- Emails via SMTP: PDF attached + download link in body.
- Inserts row in certificates + email_log.
- Updates students.status='certificate_sent'.

3.5 Build /certificates log page (Admin + Members).

3.6 Build public /verify/[certificate-id] page:

- Shows: Student name, Workshop, Date, Certificate ID,
  Status = VALID, Issued by The AI Bridge — IIUI.
- Student ID partially masked (e.g. 1234-***).

- If ID not found -> "Invalid Certificate".

DELIVERABLES:

- Reminder emails sending.
- Certificate PDFs generating with QR.
- Certificate emails sending with attachment.
- Working public verification page.

DONE WHEN:

[ ] Reminder email reaches a test student.

[ ] Certificate PDF has all fields + QR + signature.

[ ] Emailed certificate arrives with PDF attached.

[ ] Scanning QR opens verify page -> shows VALID.

[ ] Bogus certificate ID -> shows Invalid.

PHASE 4 — STUDENT PORTAL, EVENTS & PERMISSIONS

GOAL: Self-service student portal, events with photos, and admin controls.

TASKS:

4.1 Build /portal (Student Portal):

- Logged-in user -> auto-match by Google email in students table.
- If not found -> ask for Student ID -> match again.
- Show status: "In Progress (please fill post-survey)" with link,
  or "Ready" with Download Certificate button.

- Unlimited downloads.

4.2 Build /events/manage (Admin + permitted Members):

- Create / edit / delete event.
- Fields: Title, Date, Location, Short description,
  Cover image (1), Gallery images (many), SDG tags (4/8/10/17),
  Team tag (Awareness / Workshops / Analyzing),
  Visibility (public / internal).
- Upload images to Supabase Storage (event-images bucket).

4.3 Build /events (public gallery):

- Show only public events.
- Photo grid + lightbox.
- "Recent Events" strip on the homepage.

4.4 Build /admin/members (Admin only):

- List all members.
- Add/remove member.
- Toggle per-member permissions:
  can_send_reminder, can_send_cert, can_upload_event,
  can_set_visibility.
- Changes apply INSTANTLY (no re-login).

4.5 Build /admin/settings (Admin only):

- SMTP config.
- Pre/Post form URLs.
- Certificate template config (signatory name, designation).

DELIVERABLES:

- Student portal live.

- Events manager + public gallery live.
- Admin member-permission control live.
- Settings page live.

DONE WHEN:

[ ] Student logs in -> sees correct status -> downloads certificate.

[ ] Permitted member uploads an event with photos -> appears in gallery.

[ ] Internal event does NOT appear publicly.

[ ] Admin toggles a permission -> member sees change instantly.

[ ] Admin edits SMTP settings -> emails still send correctly.

PHASE 5 — POLISH, TESTING & LAUNCH

GOAL: Production-ready, tested, polished, and live before the deadline.

TASKS:

5.1 UX/UI polish pass:

- Dashboard: charts, KPI cards, clean tables, filters.
- Responsive design (mobile / tablet / desktop).
- Loading states, empty states, error toasts.
- Branding consistency from panaflex/poster/logo assets.

5.2 Security review:

- Verify RLS policies block unauthorized reads/writes.
- Verify service role key never exposed to client.
- Verify SMTP credentials stored in settings (not in code).

5.3 End-to-end testing:

- Full flow: form -> sheet -> supabase -> match -> remind ->
  cert -> email -> QR verify -> portal download.
- Edge cases: duplicate emails, missing Student ID, partial names.

5.4 Performance:

- Optimize images (Next/Image).
- Cache public pages.

5.5 Documentation:

- Admin guide (how to add members, toggle perms, upload events).
- Member guide (how to send reminders / certificates).
- Student guide (how to log in and download certificate).

5.6 Deploy to production Vercel + custom domain (if available).

5.7 Prepare demo assets for the SDG Walk stall and national SDG Gala (screenshots, live demo flow, QR code to the site).

DELIVERABLES:

- Polished production site.
- Admin / Member / Student guides.
- Demo-ready for the national exhibition.

DONE WHEN:

[ ] Full end-to-end flow passes on production.

[ ] No security holes (RLS verified).

[ ] Site is responsive and fast.

[ ] Guides written and shared with the team.

[ ] Site is live and QR code works from a phone.

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SECTION 6 — ENVIRONMENT VARIABLES (for .env.local)

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NEXT_PUBLIC_SUPABASE_URL=

NEXT_PUBLIC_SUPABASE_ANON_KEY=

SUPABASE_SERVICE_ROLE_KEY=

GOOGLE_SHEET_CSV_URL=

SMTP_HOST=

SMTP_PORT=

SMTP_USER=

SMTP_PASS=

SMTP_FROM_NAME=The AI Bridge — IIUI

SMTP_FROM_EMAIL=

CERT_SIGNATORY_NAME=Dr. Abdul Zahid Khan

CERT_SIGNATORY_DESIGNATION=Chairperson, Dept. of Technology & Project Management, FMS IIUI

CERT_SIGNATURE_IMAGE_URL=

NEXT_PUBLIC_SITE_URL=https://your-domain.com

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SECTION 7 — BUILD ORDER CHECKLIST (ONE-GLANCE)

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PHASE 1 [ ] Next.js setup [ ] Supabase setup [ ] DB schema [ ] Google Auth

[ ] Layout + branding [ ] Route protection [ ] Admin seed

PHASE 2 [ ] Publish sheet CSV [ ] Edge Function pull [ ] Cron

[ ] Matching logic [ ] Students page [ ] Dashboard

PHASE 3 [ ] SMTP config [ ] Reminder email [ ] Certificate PDF + QR

[ ] Certificate email [ ] Certificates log [ ] Verify page

PHASE 4 [ ] Student portal [ ] Events manager [ ] Public gallery

[ ] Member permissions [ ] Settings page

PHASE 5 [ ] UX polish [ ] Security review [ ] E2E testing

[ ] Docs [ ] Deploy [ ] Demo prep

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SECTION 8 — NOTES / OPEN ITEMS

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- Branding: theme will be generated after you drop panaflex, posters, and logo into the assets folder for Antigravity to analyze.
- Certificate download = unlimited (student keeps their copy).
- No manual certificate revoke (validity = form matching only).
- Google Sheet = read-only, one-way sync into Supabase.
- SMTP only for reminder + certificate emails; login is handled by Supabase Auth.
- A few dozen students expected — no scaling concerns for the pilot.

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END OF DOCUMENT

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